

Crossing-Point Analysis for Weighted Rolling Validation: Theory

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1 Motivation and setup

Rolling validation (RV) sums validation losses over successive training sizes, while the desired model has lower population risk at the *current* size. Analysis 08 compared a sample mean with an oracle that was always preferable, so it had no model-preference crossing. Analysis 09 asks whether weighting later validation times can align the expected RV crossing with a population-risk crossing while retaining reliable selection near that crossing.

Let $Y_i = \theta + \xi_i$, where the ξ_i are independent $N(0, 1)$ variables and $0 < |\theta| < 1$. Model 0 predicts with the training sample mean $\bar{Y}_n$; Model 1 always predicts zero. Both use squared loss. Write $R_k(n)$ for Model k 's population prediction risk after n training observations and define

$$\Delta_R(n) = R_0(n) - R_1(n).$$

This is a *current* risk difference, evaluated only at size n . If $RV_{k,w}(n)$ is Model k 's weighted sum of validation losses through the time with n training observations, define

$$\Delta_w(n) = \mathbb{E}[RV_{0,w}(n) - RV_{1,w}(n)].$$

This is an *accumulated expected* difference. A negative difference favors Model 0. The risk and RV calculations are in Appendices A.1, A.2, and B.1.

2 Why ordinary RV crosses late

Theorem 2.1 (Population crossing).

$$R_0(n) = 1 + \frac{1}{n}, \quad R_1(n) = 1 + \theta^2, \quad \Delta_R(n) = \frac{1}{n} - \theta^2, \quad n_{\text{pop}} = \lfloor \theta^{-2} \rfloor + 1.$$

Here n_{pop} is the first size at which $\Delta_R(n) < 0$.

Interpretation. Model 0 becomes preferable once its estimation variance falls below Model 1's squared bias. **Proof.** See Appendix A.1.

Ordinary RV gives every validation time equal weight. At time $m + 1$, the expected Model-0 minus Model-1 loss difference is $1/m - \theta^2$. Its accumulated expected difference is

$$\Delta_{\text{RV}}(n) = \sum_{m=1}^n \left(\frac{1}{m} - \theta^2 \right) = H_n - n\theta^2, \quad H_n = \sum_{m=1}^n \frac{1}{m}. \quad (1)$$

H_n is the n -th harmonic number. Population risk uses the current $1/n$ term, while RV retains earlier positive contributions. This history delays the RV crossing. The one-time expectation and sum are derived in Appendix A.2.

Proposition 2.2 (Ordinary-RV lag). *Let n_{RV} be the first size with $\Delta_{\text{RV}}(n) < 0$. As $\theta \rightarrow 0$,*

$$n_{\text{RV}} \sim \theta^{-2} \log(\theta^{-2}), \quad \frac{n_{\text{RV}}}{n_{\text{pop}}} \sim \log(\theta^{-2}) \rightarrow \infty.$$

Interpretation. The expected ordinary-RV crossing is asymptotically later than the population crossing. **Proof.** See Appendix A.3.

3 Can weighting align the crossing?

Give validation time i a positive deterministic weight w_i independent of the data. Its one-time expected Model-0 minus Model-1 loss difference is

$$d_i = \frac{1}{i-1} - \theta^2.$$

Define

$$A_n = \sum_{i=2}^{n+1} \frac{w_i}{i-1}, \quad W_n = \sum_{i=2}^{n+1} w_i.$$

Here A_n is the weighted historical estimation-variance contribution, W_n is total weight, and A_n/W_n is the weighted historical average of reciprocal training sizes. The expected weighted difference is

$$\Delta_w(n) = A_n - \theta^2 W_n. \quad (2)$$

The derivation is in Appendix B.1.

The central comparison is

$$\frac{\Delta_w(n)}{W_n} = \Delta_R(n) + \left(\frac{A_n}{W_n} - \frac{1}{n} \right). \quad (3)$$

Population risk uses $1/n$; weighted RV uses A_n/W_n . The historical correction is nonnegative, so an expected weighted-RV crossing cannot precede the population crossing (Appendix B.2). Alignment naturally calls for $A_n/W_n \sim 1/n$, equivalently $nA_n/W_n \rightarrow 1$.

Theorem 3.1 (Sufficient crossing alignment). *If the positive deterministic weights do not depend on θ and*

$$\frac{nA_n}{W_n} \rightarrow 1, \quad (4)$$

then $n_{\text{RV},w}$, the first size with $\Delta_w(n) < 0$, exists for all sufficiently small nonzero θ and

$$\frac{n_{\text{RV},w}}{n_{\text{pop}}} \rightarrow 1 \quad (\theta \rightarrow 0).$$

Interpretation. Matching the latest reciprocal training size aligns the *expected* crossings. **Proof.** See Appendix B.4.

Log, squared-log, and log-log weights reduce finite crossing lags but fail asymptotic alignment (Appendices B.5 and B.6). Geometric weights satisfy alignment by concentrating on late validation times (Appendix B.7). The weight-specific asymptotics are in those appendix sections.

4 Can alignment and concentration hold together?

A correct expected crossing is not enough: the realized weighted-RV difference

$$\Xi_{n,w} = \sum_{i=2}^{n+1} w_i \{(Y_i - \bar{Y}_{i-1})^2 - Y_i^2\}$$

is random. When its mean is nonzero, a sufficient Chebyshev concentration criterion is

$$\frac{\text{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \rightarrow 0.$$

Near the moving population crossing, $n_{\text{pop}} \sim \theta^{-2}$, so $\theta \rightarrow 0$ with $n\theta^2 \rightarrow c \in (0, \infty)$ keeps n at a fixed relative location: $c < 1$ is before the crossing and $c > 1$ is after it (Appendix C.1).

The nonzero mean in 09 introduces linear Gaussian noise

$$-2\theta \sum_{i=2}^{n+1} w_i \xi_i, \quad \text{Var}\left(-2\theta \sum_{i=2}^{n+1} w_i \xi_i\right) = 4\theta^2 \sum_{i=2}^{n+1} w_i^2.$$

This term vanishes in the $\theta = 0$ model of 08. The complete loss expansion and finite- n variance decomposition are in Appendices D.1 and D.2.

Theorem 4.1 (Near-crossing obstruction). *Under (4),*

$$n \frac{\sum_{i=2}^{n+1} w_i^2}{W_n^2} \rightarrow \infty.$$

For every sequence $\theta \rightarrow 0$ and $n\theta^2 \rightarrow c \in (0, \infty) \setminus \{1\}$,

$$\frac{\text{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \rightarrow \infty. \tag{5}$$

Interpretation. Late weighting aligns the expected crossing but reduces effective averaging. Its linear Gaussian noise prevents a vanishing Chebyshev ratio near the moving crossing. This criterion alone does not determine a limiting wrong-selection probability. **Proof.** See Appendices C.2 and C.3; the exact boundary is treated in Appendix C.4.

Log-type weights retain more averaging but remain crossing misaligned (Appendices B.5, B.6, and D.3). Geometric weights align the crossing but concentrate too strongly for near-crossing Chebyshev concentration (Appendix B.7).

What 09 establishes. Ordinary RV crosses too late because it accumulates earlier validation losses. Weighting can align the expected crossing, but under the 09 assumptions that alignment conflicts with near-crossing Chebyshev concentration.

What remains open. The result does not determine the exact limiting wrong-selection probability. Direct probability analysis or different validation designs, weight classes, or crossing models remain possible directions.

Notation used below. The observations are $Y_i = \theta + \xi_i$ with independent $\xi_i \sim N(0, 1)$ and $0 < |\theta| < 1$. Model 0 predicts with $\bar{Y}_n$, Model 1 predicts zero, and both use squared loss. For positive deterministic weights w_i , let $\Xi_{n,w} = \sum_{i=2}^{n+1} w_i \{(Y_i - \bar{Y}_{i-1})^2 - Y_i^2\}$, $\Delta_w(n) = \mathbb{E}\Xi_{n,w}$, $A_n = \sum_{i=2}^{n+1} w_i / (i-1)$, and $W_n = \sum_{i=2}^{n+1} w_i$. Also $\Delta_R(n) = R_0(n) - R_1(n)$ is the current population-risk difference, while $\Delta_{\text{RV}}(n) = \sum_{m=1}^n (1/m - \theta^2)$ is the expected ordinary-RV difference.

A Population risk and ordinary RV

A.1 Population-risk formulas and strict crossing

Let $Y_{\text{new}} = \theta + \xi_{\text{new}}$ be independent of the training sample. Since $\bar{Y}_n = \theta + S_n/n$ with $S_n = \sum_{t=1}^n \xi_t$, the Model-0 error is $\xi_{\text{new}} - S_n/n$. Independence, zero means, and $\text{Var}(S_n) = n$ give $R_0(n) = 1 + 1/n$. Model 1 has error $\theta + \xi_{\text{new}}$, hence $R_1(n) = 1 + \theta^2$. Their difference is $1/n - \theta^2$, negative exactly when $n > \theta^{-2}$. The least integer strictly above θ^{-2} is $\lfloor \theta^{-2} \rfloor + 1$, including when θ^{-2} itself is an integer (the equality index is a tie). This proves Theorem 2.1.

A.2 One-time and ordinary-RV expectation

At time $m + 1$, $\bar{Y}_m = \theta + S_m/m$ and the fresh error of Model 0 is $\xi_{m+1} - S_m/m$; that of Model 1 is $\theta + \xi_{m+1}$. Expanding and using independence of ξ_{m+1} and S_m gives

$$\mathbb{E}[(Y_{m+1} - \bar{Y}_m)^2 - Y_{m+1}^2] = (1 + 1/m) - (1 + \theta^2) = 1/m - \theta^2.$$

Summing over $m = 1, \dots, n$ proves (1).

A.3 Ordinary-RV crossing asymptotics

Put $h = \theta^2$ and $N = h^{-1}$. The mean $H_n - nh$ has increments $1/(n+1) - h$, so it is initially increasing and eventually decreasing to $-\infty$; its first strict negative index exists and is unique. Integral bounds give $\log(n+1) \leq H_n \leq 1 + \log n$. Fix $\varepsilon \in (0, 1)$ and set

$$n_- = \lfloor (1 - \varepsilon)N \log N \rfloor, \quad n_+ = \lceil (1 + \varepsilon)N \log N \rceil.$$

Both tend to infinity; $\log n_{\pm} = \log N + \log \log N + O(1)$, while $n_{\pm}h = (1 \pm \varepsilon) \log N + o(1)$. Thus $H_{n_-} - n_-h = \varepsilon \log N + o(\log N) > 0$ and $H_{n_+} - n_+h = -\varepsilon \log N + o(\log N) < 0$. The crossing lies between n_- and n_+ . Divide by $N \log N$ and let $\varepsilon \downarrow 0$ to obtain $n_{\text{RV}} \sim N \log N$. Since $n_{\text{pop}} = N + O(1)$, Proposition 2.2 follows.

B Weighted RV and crossing alignment

B.1 Weighted expectation

Apply the one-time expectation in Appendix A.2 with $m = i - 1$. Deterministic weights and linearity of expectation give

$$\Delta_w(n) = \sum_{i=2}^{n+1} w_i ((i-1)^{-1} - \theta^2) = \sum_{i=2}^{n+1} \frac{w_i}{i-1} - \theta^2 \sum_{i=2}^{n+1} w_i = A_n - \theta^2 W_n.$$

No independence between validation losses is asserted or needed.

B.2 Exact remainder and crossing order

Divide the preceding identity by $W_n > 0$ and add and subtract $1/n$ to obtain (3). Each $1/(i-1) \geq 1/n$. Their positive-weighted average therefore satisfies $A_n/W_n \geq 1/n$; for $n > 1$, at least one earlier term is strictly larger and has positive weight, so the inequality is strict. If $\Delta_R(n) \geq 0$, then $\Delta_w(n)/W_n \geq 0$. Consequently a strict weighted-RV crossing cannot precede n_{pop} . Equality is possible at $n = 1$ or if nonnegative (rather than positive) weights are supported only on the final term.

B.3 Monotonicity and crossing existence

Write $r_n = A_n/W_n$. Adding the next validation time gives

$$r_{n+1} = \frac{W_n r_n + w_{n+2}/(n+1)}{W_n + w_{n+2}}.$$

Every term in r_n is at least $1/n > 1/(n+1)$, so $r_{n+1} < r_n$ for positive weights. Thus the equation $r_n < h$ has at most one first strict crossing. It exists exactly when $\inf_n r_n < h$; since r_n decreases, this is equivalent to $\lim_n r_n < h$. If the limit equals h , strict crossing need not occur. Under (4), $r_n \sim 1/n \rightarrow 0$, so crossing exists for every fixed $h > 0$; $r_1 = 1 > h$ because $0 < h < 1$.

B.4 Sufficient-alignment theorem

Let $h = \theta^2 \downarrow 0$ and fix $\varepsilon \in (0, 1)$. Set $n_- = \lfloor (1 - \varepsilon)/h \rfloor$ and $n_+ = \lceil (1 + \varepsilon)/h \rceil$. Both grow, and (4) gives $r_{n_\pm} = (1 + o(1))/n_\pm$. Therefore

$$\frac{r_{n_-}}{h} \rightarrow \frac{1}{1 - \varepsilon} > 1, \quad \frac{r_{n_+}}{h} \rightarrow \frac{1}{1 + \varepsilon} < 1.$$

By Appendix B.3, the first strict crossing lies in $(n_-, n_+]$ for small h . Rounding changes either endpoint by at most one. Divide by h^{-1} and let $\varepsilon \downarrow 0$; since $n_{\text{pop}} = h^{-1} + O(1)$, the crossing ratio tends to one. This proves Theorem 3.1.

B.5 Log-power weights and their crossings

Let $w_i = [\log(i+1)]^p$ with fixed $p \geq 0$. A shift by one changes $\log(i+1)$ by $O(1/i)$, so its p th power has the same leading order as $(\log i)^p$ (with $p = 0$ interpreted as 1). Integral comparison, or summation by parts, yields

$$A_n \sim \int_2^n \frac{(\log x)^p}{x} dx = \frac{(\log n)^{p+1}}{p+1}.$$

For completeness, the summand is eventually decreasing: its logarithmic derivative is $(p/\log x - 1)/x < 0$ for $x > e^p$. Thus its sum differs from the integral by a bounded initial contribution and an error no larger than a fixed eventual first term. The shift error is summable for $p \geq 0$. Also, for any $\delta \in (0, 1)$, the last $(1 - \delta)n$ indices have weights $(\log n)^p(1 + o(1))$ uniformly, while the first δn contribute at most $\delta n(\log n + O(1))^p$. Taking $n \rightarrow \infty$ and then $\delta \downarrow 0$ gives $W_n \sim n(\log n)^p$. Hence

$$r_n \sim \frac{\log n}{(p+1)n}.$$

For the crossing let $N = h^{-1}$ and $g_N = N \log N/(p+1)$. At $n_\pm = \lfloor (1 \mp \varepsilon)g_N \rfloor$ and $\lceil (1 \pm \varepsilon)g_N \rceil$, respectively, $\log n_\pm / \log N \rightarrow 1$, so $r_{n_\pm}/h \rightarrow (1 \mp \varepsilon)^{-1}$. Monotonicity from Appendix B.3 brackets the first crossing between them. Letting $\varepsilon \downarrow 0$ gives $n_{\text{RV},w} \sim g_N$ and, since $n_{\text{pop}} \sim N$, the ratio stated in the main text. This includes $p = 0$.

B.6 Log-log weight

Set $f(x) = \log \log(x+e) > 0$. Its derivative is $f'(x) = 1/[(x+e)\log(x+e)]$. For large x , $f(x) \sim \log \log x$ and $f(x)/x$ is decreasing, so integral comparison applies. The shift from $f(x)$ to

$\log \log x$ produces an integrable error after division by x : the mean-value theorem bounds it by $C/(x^2 \log x)$ for sufficiently large x . Substituting $u = \log x$,

$$A_n = \int^n \frac{\log \log x}{x} dx + O(1) = \log n \log \log n - \log n + O(1) \sim \log n \log \log n.$$

The same last-fraction argument as in Appendix B.5, using $f(\delta n)/f(n) \rightarrow 1$ for fixed $\delta > 0$, gives $W_n \sim n \log \log n$. Hence $r_n \sim (\log n)/n$. Bracketing at $n_{\pm} = (1 \pm \varepsilon)N \log N$ exactly as in Appendix B.5 gives $n_{\text{RV},w} \sim N \log N$ and $n_{\text{RV},w}/n_{\text{pop}} \sim \log N$.

B.7 Geometric-weight alignment and effective averaging

For $w_i = q^i$ with $q > 1$, write $i = n + 1 - j$. Division by W_n makes the lag j have probabilities proportional to q^{-j} , $j = 0, \dots, n-1$. The denominator $\sum_{j=0}^{n-1} q^{-j} \rightarrow q/(q-1)$ and the numerator for nr_n is

$$\sum_{j=0}^{n-1} q^{-j} \frac{n}{n-j}.$$

For $j \leq n/2$, $n/(n-j) \leq 2$ and tends to 1 for each fixed j ; dominated convergence for the summable geometric series gives the same limit $q/(q-1)$. For $j > n/2$, the sum is at most $n \sum_{j>n/2} q^{-j} \rightarrow 0$. Thus $nr_n \rightarrow 1$. Direct finite geometric sums also give

$$\frac{\sum_{i=2}^{n+1} q^{2i}}{(\sum_{i=2}^{n+1} q^i)^2} \rightarrow \frac{q-1}{q+1}.$$

Theorem 3.1 supplies the crossing ratio; the displayed square-weight limit measures its bounded effective averaging.

C Near-crossing alignment–concentration obstruction

C.1 The near-crossing scaling

Put $h = \theta^2$. Theorem 2.1 gives $n_{\text{pop}} = h^{-1} + O(1)$. Thus $nh \rightarrow c$ places n at asymptotic fraction c of the population crossing. Moreover $n\Delta_R(n) = 1 - nh \rightarrow 1 - c$, proving the before/after sign interpretation for $c \neq 1$.

C.2 Weight concentration and effective sample size

Let $p_{n,i} = w_i/W_n$, so $\sum_i p_{n,i} = 1$. Alignment is

$$\sum_{i=2}^{n+1} p_{n,i} \left(\frac{n}{i-1} - 1 \right) \rightarrow 0.$$

All summands are nonnegative. For fixed $\varepsilon \in (0, 1)$ and $i-1 \leq (1-\varepsilon)n$, the factor is at least $\varepsilon/(1-\varepsilon)$. Hence the total mass there is at most $(1-\varepsilon)/\varepsilon$ times the displayed sum, and tends to zero. The remaining set has at most $\varepsilon n + 1$ indices and mass tending to one. Cauchy–Schwarz on that set gives

$$\sum_i p_{n,i}^2 \geq \frac{(\sum_{i-1 > (1-\varepsilon)n} p_{n,i})^2}{\varepsilon n + 1}, \quad \text{so} \quad \liminf_n n \sum_i p_{n,i}^2 \geq \frac{1}{\varepsilon}.$$

For any $K > 0$, choose $\varepsilon < 1/K$; the lower limit is then at least K . Thus $n \sum_i p_{n,i}^2 \rightarrow \infty$.

C.3 Proof of the obstruction theorem

Under alignment, $r_n = (1 + o(1))/n$. If $nh \rightarrow c \neq 1$, then by (2)

$$\frac{\mathbb{E}\Xi_{n,w}}{W_n} = r_n - h = \frac{1 - c + o(1)}{n}.$$

Appendix D.2 proves that the quadratic variance is nonnegative and the additional linear variance equals $4h \sum_i w_i^2$. Therefore

$$\frac{\text{Var}(\Xi_{n,w})}{(\mathbb{E}\Xi_{n,w})^2} \geq \frac{4h \sum_i p_{n,i}^2}{(r_n - h)^2} = \frac{4(c + o(1)) n \sum_i p_{n,i}^2}{(1 - c + o(1))^2} \rightarrow \infty$$

by Appendix C.2. The mean is positive when $c < 1$ and negative when $c > 1$, for sufficiently large n . The proof establishes failure of a sufficient Chebyshev argument, not a positive lower bound on wrong-selection probability. This proves Theorem 4.1.

C.4 The exact boundary

If $nh \rightarrow 1$ under alignment, $r_n - h = o(1/n)$. When this difference is zero, the variance-to-mean-squared ratio is undefined. When it is nonzero, the preceding lower bound has numerator $4h \sum p_{n,i}^2$ and denominator $o(n^{-2})$; writing $r_n - h = \delta_n/n$ with $\delta_n \rightarrow 0$ gives a lower bound $4(1 + o(1))n \sum p_{n,i}^2/\delta_n^2 \rightarrow \infty$. No leading-order population sign is assigned at $c = 1$.

D Exact variance and a fixed-parameter comparison

D.1 Loss expansion and centering

Write $S_m = \sum_{t=1}^m \xi_t$. Since $\bar{Y}_m = \theta + S_m/m$,

$$(Y_{m+1} - \bar{Y}_m)^2 - Y_{m+1}^2 = (\xi_{m+1} - S_m/m)^2 - (\theta + \xi_{m+1})^2 = \frac{S_m^2}{m^2} - 2\xi_{m+1} \frac{S_m}{m} - 2\theta\xi_{m+1} - \theta^2.$$

The first quadratic term has mean $1/m$ and the other random terms have mean zero. Therefore, with

$$Q_{n,w} = \sum_{m=1}^n w_{m+1} \left\{ \frac{S_m^2}{m^2} - 2\xi_{m+1} \frac{S_m}{m} \right\},$$

we have $\Xi_{n,w} - \mathbb{E}\Xi_{n,w} = Q_{n,w} - 2\theta \sum_{i=2}^{n+1} w_i \xi_i$. This also independently verifies (2).

D.2 Gaussian orthogonality and coefficients

Expand $S_m^2 - m = \sum_{t=1}^m (\xi_t^2 - 1) + 2 \sum_{1 \leq t < u \leq m} \xi_t \xi_u$. Collecting each centered square gives coefficient

$$a_t = \sum_{m=t}^n \frac{w_{m+1}}{m^2}, \quad 1 \leq t \leq n.$$

For each pair $t < l$, the square expansion contributes $2 \sum_{m=l}^n w_{m+1}/m^2$; the validation term at $m = l - 1$ contributes $-2w_l/(l - 1)$. Thus, with empty sums zero,

$$b_l = \sum_{m=l}^n \frac{w_{m+1}}{m^2} - \frac{w_l}{l - 1}, \quad 2 \leq l \leq n + 1,$$

and

$$Q_{n,w} = \sum_{t=1}^n a_t(\xi_t^2 - 1) + 2 \sum_{l=2}^{n+1} b_l \sum_{t=1}^{l-1} \xi_t \xi_l.$$

Independent standard Gaussians satisfy $\mathbb{E}(\xi_t^2 - 1)^2 = 2$ and $\mathbb{E}(\xi_t \xi_l)^2 = 1$ for $t \neq l$. Distinct centered squares, distinct unordered pair products, and a centered square against a pair product have covariance zero: in every cross-product, some independent centered Gaussian appears to an odd power or as a mean-zero factor. Each quadratic monomial also has zero covariance with every ξ_i , because every centered third Gaussian moment is zero. Hence the quadratic and linear parts are orthogonal, and

$$\text{Var}(Q_{n,w}) = 2 \sum_{t=1}^n a_t^2 + 4 \sum_{l=2}^{n+1} (l-1) b_l^2, \quad \text{Var}\left(-2\theta \sum_{i=2}^{n+1} w_i \xi_i\right) = 4\theta^2 \sum_{i=2}^{n+1} w_i^2.$$

Adding proves the exact finite- n variance identity

$$\text{Var}(\Xi_{n,w}) = 2 \sum_{t=1}^n a_t^2 + 4 \sum_{l=2}^{n+1} (l-1) b_l^2 + 4\theta^2 \sum_{i=2}^{n+1} w_i^2. \quad (6)$$

D.3 Slow-weight variance orders

Here $h = \theta^2$ and $nh \rightarrow c \in (0, \infty)$; consider $w_{m+1} = f(m)$, where $f(m) = 1, [\log(m+2)]^p$ for fixed $p > 0$, or $\log \log(m+1+e)$. In each case $f(n)$ is slowly varying, and splitting sums at δn as in Appendix B shows

$$\sum_{m=1}^n f(m)^2 \sim n f(n)^2.$$

We verify the quadratic order without borrowing it from 08. The coefficients a_t are bounded by $\sum_{m=t}^{\infty} f(m)/m^2$. For the listed f , integral comparison and eventual monotonicity of $f(x)/x^{3/4}$ give $\sum_{m=t}^{\infty} f(m)/m^2 \leq C f(t)/t$ for large t . Consequently $\sum_t a_t^2 \leq C + C \sum_t f(t)^2/t^2 = O(1)$.

Set $T_n = \sum_{m=n+1}^{\infty} f(m)/m^2$ and $B_l = \sum_{m=l}^{\infty} f(m)/m^2 - f(l-1)/(l-1)$, so $b_l = B_l - T_n$ for $2 \leq l \leq n$ and $b_{n+1} = -f(n)/n$ (the harmless one-step indexing changes the following estimates only by lower-order terms). Integral comparison, summation by parts, and the mean-value theorem give

$$T_n \sim f(n)/n, \quad |B_l| \leq C \left(\frac{\sup_{x \in [l-1, l+1]} |f'(x)|}{1} + \frac{f(l)}{l^2} \right)$$

for smooth extensions of the listed f . To see the derivative bound, subtract the constant- $f(l)$ sum, whose discrepancy from $f(l)/(l-1)$ is $O(f(l)/l^2)$; summation by parts bounds the remaining variation by $C \int_l^{\infty} |f'(x)| x^{-1} dx$. This is $O((\log l)^{p-1}/l)$ for log powers and $O(1/(l \log l))$ for log-log; the constant case has zero derivative. Therefore $D_n = \sum_{l=2}^n (l-1) B_l^2$ is $O(1)$ for $f = 1$ and log-log, while for log powers it is bounded by

$$C + C \sum_{l=3}^n \frac{(\log l)^{2p-2}}{l} = \begin{cases} O(1), & 0 < p < 1/2, \\ O(\log \log n), & p = 1/2, \\ O((\log n)^{2p-1}), & p > 1/2. \end{cases}$$

The three cases follow by $u = \log x$ in the comparison integral $\int^n (\log x)^{2p-2} dx/x$. In every case $D_n = o(f(n)^2)$ when $f(n) \rightarrow \infty$; for $f = 1$, $D_n = O(1)$ and the exact $B_l = O(l^{-2})$ is handled

below. The terminal contribution is

$$4T_n^2 \sum_{l=2}^n (l-1) \sim 4 \frac{f(n)^2 n^2}{n^2} \frac{n^2}{2} = 2f(n)^2.$$

Expanding $(B_l - T_n)^2$, Cauchy–Schwarz bounds the magnitude of its cross term by $2\{D_n T_n^2 \sum_{l=2}^n (l-1)\}^{1/2} = o(f(n)^2)$ for the growing weights. The end coefficient contributes $4n(f(n)/n)^2 = o(f(n)^2)$. Thus $\text{Var}(Q_{n,w}) \sim 2f(n)^2$ for the growing log and log-log weights. For $f = 1$, direct integral comparison yields $B_l = O(l^{-2})$. Thus $D_n = O(1)$ and $\sum_{l=2}^n (l-1)|B_l| = O(\log n)$. Since $T_n = O(1/n)$, the cross term is $O(\log n/n) = o(1)$; the end coefficient is $O(1/n)$. Therefore $\text{Var}(Q_{n,w}) \sim 2$ also in the unweighted case.

The linear variance is $4h \sum_i w_i^2 \sim 4cf(n)^2$. Hence the full variance is of order $f(n)^2$ (indeed asymptotic to $(2 + 4c)f(n)^2$ for the growing weights). Meanwhile Appendix B gives $\mathbb{E}\Xi_{n,w} \sim (\log n)f(n)/(p+1)$ for log powers and $\sim (\log n)f(n)$ for log-log; the $-hW_n$ term is only $O(f(n))$. The ordinary-weight mean is $\sim \log n$ and its full variance tends to $2 + 4c$. Therefore all four Chebyshev ratios have order $(\log n)^{-2}$. Their means are positive for fixed c , establishing the sign statement in the main text.

D.4 Stretched-exponential weights at fixed θ

Let $w_i = \exp(i^\beta)$ with $0 < \beta < 1$. Put $F(x) = e^{x^\beta}$. Substitution $u = x^\beta$ and integration by parts yield, for $a = 1, 2$,

$$\int_0^n F(x)^a dx = \frac{1}{\beta} \int_0^{n^\beta} e^{au} u^{1/\beta-1} du \sim \frac{n^{1-\beta} e^{an^\beta}}{a\beta}.$$

Because F is increasing, bounding sums between adjacent integrals changes them by at most $O(F(n)^a)$, negligible relative to $n^{1-\beta}F(n)^a$. Endpoint shifts by one are negligible since $(n+1)^\beta - n^\beta \rightarrow 0$. Therefore

$$W_n \sim \frac{n^{1-\beta} e^{n^\beta}}{\beta}, \quad \sum_i w_i^2 \sim \frac{n^{1-\beta} e^{2n^\beta}}{2\beta}, \quad \frac{\sum_i w_i^2}{W_n^2} \sim \frac{\beta}{2n^{1-\beta}} \rightarrow 0.$$

These sums show an effective terminal window of order $n^{1-\beta} = o(n)$. For any fixed $\delta > 0$, the total weight at $i - 1 \leq (1 - \delta)n$ is at most $ne^{((1-\delta)n + O(1))^\beta} = o(W_n)$, because the exponential gap is of order n^β . On the remaining indices, $n/(i-1) \leq 1/(1-\delta)$; for the earlier indices their weighted contribution to nr_n is negligible by the same exponential gap even after the factor $n/(i-1) \leq n$. Hence $1 \leq \liminf nr_n \leq \limsup nr_n \leq 1/(1-\delta)$, and $\delta \downarrow 0$ proves alignment.

For fixed $\theta \neq 0$, $\mathbb{E}\Xi_{n,w}/W_n = r_n - \theta^2 \rightarrow -\theta^2$. To bound the quadratic variance, define the centered single-time quadratic increment $U_m = (S_m^2 - m)/m^2 - 2\xi_{m+1}S_m/m$. Gaussian moments and independence give $\|U_m\|_2 \leq \sqrt{2}/m + 2/\sqrt{m} \leq C/\sqrt{m}$. Minkowski's inequality and then Cauchy–Schwarz yield

$$\left\| \frac{Q_{n,w}}{W_n} \right\|_2 \leq C \sum_{m=1}^n \frac{w_{m+1}}{W_n \sqrt{m}} \leq C \left(\sum_{m=1}^n \frac{w_{m+1}}{W_n m} \right)^{1/2} = C \sqrt{r_n} \rightarrow 0.$$

The normalized linear variance is $4\theta^2 \sum_i w_i^2/W_n^2 \rightarrow 0$. Orthogonality from Appendix D.2 then gives $\text{Var}(\Xi_{n,w})/W_n^2 \rightarrow 0$, and division by the mean-squared limit θ^4 proves fixed- θ concentration. This does not apply to $\theta \rightarrow 0$ with $n\theta^2 \rightarrow c$.